

PAPER_762: NGC 1792 The Stellar Forge — UQFF Starburst Galaxy Evolution

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Framework: UQFF (Universal Quantum Field Superconductive Framework)

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Abstract

NGC 1792 (“The Stellar Forge”) is a starburst spiral galaxy located ~42 million light-years away in Columba, forming stars over $10\times$ faster than the Milky Way at $\sim 10\text{ M}\odot/\text{yr}$. This paper derives the Master Universal Gravity UQFF equation governing its starburst-driven evolution, incorporating stellar mass growth, supernova feedback suppression, cosmic expansion, and Aether electromagnetic effects. The result $g_{\text{NGC1792}} \approx 1.053\times 10^{-2}\text{ m/s}^2$ is dominated by the $[\text{UA}]/[\text{SCm}]$ electromagnetic Aether term.

1. Introduction

NGC 1792 is a classic starburst spiral galaxy with a bright orange core of older stars surrounded by blue young star-forming regions. Its high SFR ($\sim 10\text{ M}\odot/\text{yr}$) is driven by a large gas reservoir, potentially triggered by past tidal interactions. Supernova feedback from rapid star formation injects energy into the ISM, regulating further star formation over $\sim 100\text{ Myr}$ timescales. The UQFF framework captures these dynamics through non-standard Aether and superconductive magnetism terms.

2. Master UQFF Gravity Equation

$$g_{\text{NGC1792}}(r, t) = (G * M) / r^2 * (1 + H(z)*t) * (1 + M_{\text{sf}}(t)) * (1 - F_{\text{sn}}(t)) * (1 + f_{\text{TRZ}}) + q*(v \times B) * (1 + \rho_{\text{vac},[\text{UA}] / \rho_{\text{vac},[\text{SCm}])} * 10^{-12}$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	$10^{10}\text{ M}\odot = 1.989\times 10^{40}\text{ kg}$	Hubble
Galaxy radius ($\frac{1}{2}$ diam.)	r	$3.78\times 10^{20}\text{ m}$ ($\sim 40\text{ kly}$)	Hubble
Redshift	z	0.0095	Distance calc
Starburst duration	t	$100\times 10^6\text{ yr} = 3.156\times 10^{15}\text{ s}$	Labs
Star formation rate	SFR	$10\text{ M}\odot/\text{yr}$	Hubble
Feedback amplitude	F_0	0.05	Labs estimate
Feedback timescale	τ_{sn}	$100\times 10^6\text{ yr} = 3.156\times 10^{15}\text{ s}$	Labs
Orbital velocity	v	10^6 m/s	ISM
Galactic B field	B	10^{-5} T	Labs
$\rho_{\text{vac},[\text{UA}]}$	—	$7.09\times 10^{-36}\text{ J/m}^3$	UQFF
$\rho_{\text{vac},[\text{SCm}]}$	—	$7.09\times 10^{-37}\text{ J/m}^3$	UQFF
f_{TRZ}	—	0.1	UQFF

3. Long-Form Derivation

Step 1: Base Gravitational Term

$$g_{\text{grav}} = (6.6743\text{e-}11 \times 1.989\text{e}40) / (3.78\text{e}20)^2 \\ = 1.328\text{e}30 / 1.429\text{e}41 = 9.293\text{e-}12 \text{ m/s}^2$$

Step 2: Starburst Mass Growth

$$M_{\text{sf}}(t) = \text{SFR} \times t / M_0 = 10 \times 100\text{e}6 / 10^{10} = 0.1 \\ 1 + M_{\text{sf}}(t) = 1.1$$

Step 3: Supernova Feedback Adjustment

$$t / \tau_{\text{sn}} = 3.156\text{e}15 / 3.156\text{e}15 = 1 \\ F_{\text{sn}}(t) = 0.05 \times (1 - \exp(-1)) = 0.05 \times 0.6321 = 0.031605 \\ 1 - F_{\text{sn}}(t) = 0.968395$$

Step 4: Cosmic Expansion

$$H(z) = 70 \times \sqrt{0.3 \times (1.0095)^3 + 0.7} = 70 \times \sqrt{1.00861} = 70.301 \text{ km/s/Mpc} \\ H(z) = 70.301\text{e}3 / 3.086\text{e}22 = 2.278\text{e-}18 \text{ s}^{-1} \\ H(z) \times t = 2.278\text{e-}18 \times 3.156\text{e}15 = 7.189\text{e-}3 \\ 1 + H(z) \times t = 1.007189$$

Step 5: Time-Reversal Correction

$$1 + f_{\text{TRZ}} = 1.1$$

Step 6: Electromagnetic [UA] Term

$$q \times (v \times B) = 1.602\text{e-}19 \times 1\text{e}6 \times 1\text{e-}5 = 1.602\text{e-}18 \text{ N} \\ a = 1.602\text{e-}18 / 1.673\text{e-}27 = 9.575\text{e}8 \text{ m/s}^2 \\ (1 + \rho_{\text{vac}}[\text{UA}] / \rho_{\text{vac}}[\text{SCm}]) = 11 \\ \text{Total} = 9.575\text{e}8 \times 11 \times 10^{-12} = 1.053\text{e-}2 \text{ m/s}^2$$

Step 7: Final Solution

$$g_{\text{NGC1792}} = (9.293\text{e-}12) \times (1.007189) \times (1.1) \times (0.968395) \times (1.1) + 1.053\text{e-}2 \\ = 1.097\text{e-}11 + 1.053\text{e-}2 \\ \approx 1.053\text{e-}2 \text{ m/s}^2$$

4. Physical Interpretation

The starburst phase dominates NGC 1792's evolution. The supernova feedback term ($1 - F_{\text{sn}} = 0.968$) captures feedback self-regulation, while the mass growth term ($1 + M_{\text{sf}} = 1.1$) reflects the 10% gas-to-star conversion rate over 100 Myr. The Aether [UA] electromagnetic term ($1.053 \times 10^{-2} \text{ m/s}^2$) exceeds classical gravity by ~6 orders of magnitude under UQFF, revealing non-standard vacuum energy as a primary driver.

5. UQFF Framework Advancement

- First UQFF application to a spiral starburst galaxy at 42 Mly
 - Supernova feedback modeled as a damping term on effective buoyancy
 - Starburst star formation mass growth incorporated as additive UQFF correction
 - Demonstrates UQFF scalability to galaxy-scale starburst environments
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6. Conclusions

The Master UQFF gravity equation for NGC 1792 yields $g_{\text{NGC1792}} \approx 1.053 \times 10^{-2} \text{ m/s}^2$, confirming the Aether electromagnetic term dominates starburst galaxy dynamics under UQFF. The result encodes the balance between starburst mass growth and supernova feedback suppression over a 100 Myr evolutionary timescale.

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